

NETWORKED CONTROL

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EXERCIZES

1) Consider the following continuous-time system $\mathcal{S}$.

$$\dot{x}(t) = (\log_e 2)x(t) + \log_e 8 u(t)$$

- A. Compute the eigenvalue and the mode of the system. Is the system open-loop asymptotically stable?
- B. We aim to control the system using a digital quantized controller. Answer the following questions.
 - a. According to the theory, what is the minimum rate N_B/h (where N_B is the number of bits encoding the control law and h is the sampling time) required to make the system *boundable*?
 - b. In view of the answer given to the previous question, what is the maximum value of h that can be used to make the system *boundable* in case $N_B = 2$?
 - c. Set $h = 2$ s and $N_B = 2$. Using the control law defined in case we operate at the rate limits (with $u_{\min} = -1$ and $u_{\max} = 1$), analyze the properties of the state trajectories using the graphical method (possibly starting by different selected initial conditions).
 - d. Is the control law used at point c. the unique one that allows to steer the state to lie in a bounded set? Why? Compute the corresponding invariant set.

2) Assume that the linear system

$$\mathcal{S}: \begin{cases} \dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) \end{cases}$$

is controlled through the networked control system (NCS) sketched in the following figure.

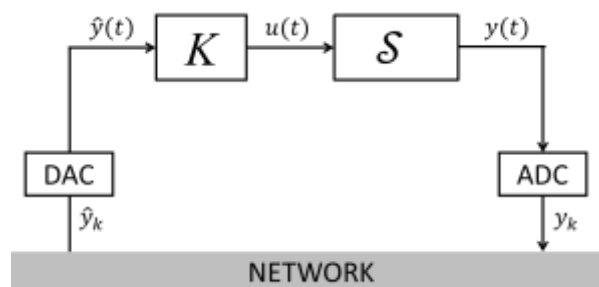


Figure 1: Networked control system

Assume that the NCS is subject to a time-invariant delay τ and to packet dropouts.

- A. Define the asymptotic dropout rate r .
- B. Define the maximal dropout rate and state, in details, the main result allowing to compute it, properly defining all the necessary ingredients.

3) How, qualitatively, the choice of the sampling time affects the performance of a networked control system, in case delays and packet dropouts affect the control network?